

Expenses Reporting Tool for an IT Consultancy

Project Management (GEP)
Assignment 4: Final Document

Bachelor Degree in Informatics Engineering
Specialization in Software Engineering

Bachelor's Thesis



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5 Temporal Planning

This section introduces the initial **122-day** project plan, scheduled from February 18, 2026, to June 19, 2026. The total estimated effort for all defined tasks is **513 hours**.

According to Polytechnic University of Catalonia, the bachelor's thesis carries a workload of 18 ECTS credits, equivalent to 540 effort hours. The discrepancy between the total estimated effort of 513 hours and the expected 540 hours results in a built-in buffer of **27 hours** to accommodate potential setbacks.

5.1 Task Description

The key to achieving effective time planning lies in a thorough analysis of all tasks, their priorities and durations. This step facilitates proper task sequencing and resource allocation. For this reason, all tasks are outlined in this section and categorized into six main groups: project starting, project management, final project delivery, design, development, and frontend/backend development.

5.1.1 Project Starting

A solid project start requires a preliminary study and initial project definition. This aligns the project vision among all involved members and establishes a unified understanding of the project's scope and objectives. To achieve this, the subsequent tasks are defined:

PS1. Context and Scope

This task is divided into smaller sub-tasks:

- Analyze the current problem and identify all stakeholders.
- Define and propose a technical solution.
- Describe the project context and conduct a market study.
- Define the project scope in terms of objectives, requirements, obstacles, and risks.
- Select a pattern methodology to guide the entire project development.
- Select the monitoring and validation tools to be used in the project.

PS2. Temporal Planning

Identify all tasks and resources, design an initial Gantt diagram timeline, and define a risk management strategy.

PS3. Budget and Sustainability

Define a project budget and analyze the sustainability aspects of the proposed solution.

PS4. Initial Documentation

Consolidate the documentation from previous tasks into the definitive version of the initial documentation.

5.1.2 Project Management

Project management is essential for monitoring the project's progress and keeping the project on track.

PM1. Agile Ceremonies

Run daily scrums, sprint plannings, sprint reviews, and sprint retrospectives.

PM2. Thesis Director Review Meetings

Meet with the thesis director every two weeks to ensure the project is progressing correctly.

PM3. Product Backlog Update

Refine, update, and maintain the product backlog, which always reflects the current project status.

5.1.3 Final Project Delivery

Tasks for completing product delivery are listed below.

FPD1. User Manual Documentation

Write a user manual for the application's end-users as a reference guide.

FPD2. Final Thesis Documentation

Write the bachelor's thesis final documentation intended for an academic audience. This task involves structuring the full project development, from planning and design to results and conclusions, maintaining an academic tone and standard.

FPD3. Final Demonstration Video

Record a final demonstration of the application once it is ready for use.

FPD4. Bachelor's Thesis Defense

Plan and prepare a final exposition to defend and justify the bachelor's thesis achievements, while demonstrating commitment to the academic standards.

5.1.4 Design

This preliminary design phase establishes a clear application architecture to guide the subsequent development phase.

DT1. Use Case Diagram

Analyze all users' interactions with the system and design a Use Case Diagram that illustrates all actors, use cases, and relationships between use cases.

DT2. Conceptual Model Diagram

Design the UML database diagram, clarifying all database entities, their attributes, and the relationships between them.

DT3. Technology Stack Decision

Select the technology to be used for each application layer and provide a clear technology selection documentation that serves as a guide for the application development phase.

DT4. Logical and Physical Architecture Design

Design the logical and physical application architecture.

The former describes how the implementation code is structured, including the purpose and role of each application layer. The latter explains how the application connects to external services and resources, both physical and virtual, and describes the function of each.

DT5. UI mockups Design

Deliver the mockup designs for each application page.

DT6. Security Design

Provide a security architecture design for user authentication, information submission and storage, and access control.

5.1.5 Development

Essential tasks for the application development are specified in this task group.

Dev0. Self-learning and study of unfamiliar technologies

Study and learn new application technologies in advance.

Dev1. Environment Configuration

Configure a ready-to-use environment to begin application development. This task involves installing Visual Studio Code, essential extensions, creating a Git repository, and similar setup tasks.

Dev2. Product Deployment

Release the final product for end-user availability.

Dev3. Unit and Integration Testing

Include testing to verify that the implemented code functions correctly and to prevent coding bugs.

Dev4. Logging and Monitoring

Set up logging and monitoring tools to track application performance and review logs.

Dev5. Git Repository Documentation

Maintain clear repository documentation, including an application setup guide, API contracts documentation, and meaningful code comments. This documentation is targeted to a technical audience.

5.1.6 Frontend and Backend Development

As a full-stack web application, the implementation is divided into two main areas. The frontend is responsible for the user interface (UI) and user experience (UX). The backend implements the complex implementations hidden in the frontend via a RESTful API and integrates external services.

This development stage requires both frontend and backend development. Since it covers a wide range of functionalities, it is divided into two subgroups: Expense and Report Development, and User Access Development.

Expense and Report Development

This section includes all tasks related to the application's core functionalities, namely the implementation of the expense and report features.

ER1. Expense Uploads

Develop a file submission feature where the frontend handles the expense drag-and-drop and upload interface, and the backend registers the uploaded expense in the database.

ER2. Uploaded Expenses Real-time Dashboard

Develop a visual dashboard to display all uploaded expenses with their real-time processing status, which can be:

- **Processing:** The expense is pending processing by the AI model.
- **Pending Review:** The AI model has successfully analyzed the expense, the uploader can now review it.
- **Completed:** The user has reviewed and completed all required information. The expense is ready for being included in a new or existing report.
- **Failed:** The AI couldn't extract data from the receipt (due to unavailability or unrecognized format).

ER3. AI-Extracted Expense Data Review and Correction

Develop a form page for users to review and edit AI-processed expenses and fill in missing required information. Once the expense information is saved, the database is updated with the corrected data.

ER4. Uploaded Expenses Deletion

Develop the expense deletion feature to delete one or multiple expenses at once, regardless of their current status. When the user confirms deletion, the respective expenses are marked as deleted in the database.

ER5. Expense Reports View, Creation, Edit, Submission, and Deletion

Develop report management functionalities (create, edit, submit, and delete reports) and the required RESTful API for synchronizing the database with the frontend changes.

User Access Development

The tasks described in this subgroup are contingent upon user authentication implementation before their execution can begin. It is important to note that administrators can set spending limits and manage and view submitted reports solely from employees under their supervision.

UA1. Company Email User Authentication

Develop the login feature that allows users to authenticate in the application through the company's email account:

1. Verify the user access.
2. (a) On the first login, create a new user account.
(b) On the return visit, recover the relevant data of the corresponding user.

UA2. Role-based User Access

Develop the user role assignment (based on two main roles: regular employee or administrator) and expose only the corresponding role's functionalities.

UA3. Submitted Report Approval and Rejection

Develop a feature to allow administrators to review, approve, or reject submitted reports, and update the database accordingly.

UA4. Employee Spending Limit Configuration

Develop an admin-only interface to set employees' spending limits and record the changes in the database.

UA5. Submitted and Reviewed Expenses Records Visualization

Develop a dashboard displaying the records of both users' submitted reports and administrators' reviewed reports.

UA6. Email Notification Service Integration

Develop the email notification service to notify users when their submitted reports are reviewed, and to administrators when new reports are pending for their review.

5.1.7 Backend Development

The following tasks are implemented exclusively in the application backend logic.

BD1. RESTful API Backend Logic Implementation

Configure the backend architecture defined in the DT3 task and build a RESTful API to manage backend responsibilities through frontend requests.

BD2. Azure SQL Database Integration

Integrate Azure SQL Database with Entity Framework Core migrations to manage users' relevant information (e.g., extracted expense and report information, employees' hierarchy) and configure the initial database according to the diagram designed in DT2 task.

BD3. Azure Blob Storage Integration

Store original expense files using Azure Blob Storage.

BD4. Microsoft Document Intelligence AI Model Integration

Automate the expense data extraction through the Microsoft Document Intelligence AI model integration.

BD5. Automated AI-driven Data Extraction Pipeline

Integrate Event Grid, Service Bus, and Azure Function and implement a mechanized data extraction workflow that prevents bottlenecks during multiple simultaneous uploads:

1. Save the uploaded receipt file in the Blob Storage and store the receipt information in the SQL Database.
2. Detect the event using Azure Event Grid and route the expense file to the Service Bus queue.

3. Once the new expense is in the queue, trigger the Azure Function to process the receipt file using the integrated AI model.

BD6. Azure SignalR Real-time Notification Service Integration

Integrate SignalR to configure a SignalR Hub that sends real-time expense status updates, eliminating manual page refreshing whenever the AI processes a new receipt file.

BD7. Microsoft Entra ID User Authentication Integration

Handle employee authentication through the company email account by using Microsoft Entra ID integration.

5.1.8 Frontend Development

This section covers all tasks that require only frontend logic implementation.

FD1. Expenses Search and Filter

Develop visual tools that enable users to search uploaded expenses by name or filter them by category.

FD2. Reports Search

Provide a search bar to search specific reports by name.

5.2 Resources

Following the analysis of all required tasks for successful project completion, all necessary resources must be identified. Resources are divided into two main groups: human resources and physical resources. All human resources are listed in Table 5.2.1, specifying the responsible individual for each role. The physical resources are further categorized into software and hardware resources, and are identified and briefly described in Tables 5.2.2 and 5.2.3.

5.2.1 Human Resources

Resource ID	Role Name	Responsible Individual
SM	Scrum Master	Thesis author
PO	Product Owner	Sergio Alejandro Lobig Machado
TD	Thesis Director	Marc Oriol Hilari
GT	GEP Tutor	Eduard Balbuena Longo
DT	Development Team	Thesis author

Table 5.2.1: Human resources. [Own creation]

5.2.2 Physical Resources

Hardware Resources

Resource ID	Resource Name	Components
OS	Office Supplies	A table, a chair, electricity, internet, and office rental.
HW	Hardware Resources	A computer, two monitors, a mouse, and a keyboard.

Table 5.2.2: Hardware resources. [Own creation]

Software Resources

Resource ID	Resource Name	Description
GIT	Git	A version control tool for managing the source code repository.
AZR	Microsoft Azure Resources	Cloud services and resources: SQL Database, Blob Storage, AI model, Event Grid, Azure Function, Service Bus, SignalR, EntraID, and Azure DevOps (for repository and product backlog).
MCS	Microsoft Collaboration Tools	Communication and coordination platforms: Microsoft Teams, Calendar, Outlook, and Word.
VSC	Visual Studio Code	Source code editor with extensions and debugging tools.
FIB	FIB Platforms	Platforms from Barcelona School of Informatics(FIB): Racó, Atenea, FIB secretary, bachelor's thesis official website guide.
GM	Gmail	Email communication with university entities.
WEB	Web Browsers and Search Engines	Research Browsers: Mozilla Firefox, Google Chrome, Microsoft Edge.
OVL	Overleaf LaTeX Editor	Thesis formatting editor for final thesis documentation.
OT	Online Tools	Diagram design tools: Edraw, Draw.io, and Gantt Project.

Table 5.2.3: Software resources. [Own creation]

5.3 Summary of Tasks and Resources

See Table 5.3.1 for a summary of all defined tasks, including their identifiers, estimated effort in hours, any final-to-start dependencies with other tasks, and the required resources. The final-to-start dependency column indicates which task must be fully completed before the task in that row can begin.

ID	Name	Time (h)	Dependencies	Resources
PS	Project Starting	50		
PS1	Context and Scope	20	-	SM, PO, TD, GT, DT, OS, HW, MCS, VSC, FIB, GM, WEB, OVL
PS2	Temporal Planning	10	PS1	SM, PO, TD, GT, DT, OS, HW, MCS, VSC, FIB, GM, WEB, OVL, OT
PS3	Budget and Sustainability	10	PS2	SM, PO, TD, GT, DT, OS, HW, MCS, VSC, FIB, GM, WEB, OVL
PS4	Initial Documentation	10	PS3	SM, PO, TD, GT, DT, OS, HW, MCS, VSC, FIB, GM, WEB, OVL
PM	Project Management	67		
PM1	Agile Ceremonies	51	PS2	SM, PO, DT, OS, HW, AZR, MCS, VSC
PM2	Thesis Director Review Meetings	8	-	DT, TD, OS, HW, AZR, VSC, FIB, GM, WEB
PM3	Product Backlog Update	8	PS2	SM, DT, PO, OS, HW, AZR, WEB
FD	Final Product Delivery	46		
FPD1	User Manual Documentation	2	PM5	DT, OS, HW, VSC, WEB
FPD2	Final Thesis Documentation	20	-	DT, TD, OS, HW, FIB, GM, WEB, OVL, OT
FPD3	Final Demonstration Video	4	FPD2	DT, OS, HW, WEB
FPD4	Bachelor's Thesis Defense	20	FPD2	DT, TD, OS, HW, FIB, GM, WEB
DT	Design	16		
DT1	Use Case Diagram	2	PS1	DT, OS, HW, OT, WEB, MCS
DT2	Conceptual Model Diagram	4	PS1	DT, OS, HW, OT, WEB
DT3	Technology Stack Decision	2	PS1	DT, OS, HW, WEB, MCS
DT4	Logical and Physical Architecture Design	4	PS1	DT, OS, HW, WEB, MCS
DT5	UI mockups Design	2	DT1	DT, OS, HW, OT, WEB
DT6	Security Design	2	DT3, DT4	DT, OS, HW, WEB, MCS
Dev	Development	76		
Dev0	Self-learning and study of unfamiliar technologies	8	DT3	DT, OS, HW, WEB
Dev1	Environment Configuration	2	DT2, DT5, DT6	DT, OS, HW, GIT, AZR, VSC, WEB
Dev2	Product Deployment	8	Dev3	DT, OS, HW, GIT, AZR, VSC, WEB
Dev3	Unit and Integration Testing	10	UA4, UA5, UA6	DT, OS, HW, GIT, AZR, VSC, WEB
Dev4	Logging and Monitoring	8	Dev2	DT, OS, HW, GIT, AZR, VSC, WEB
Dev5	Git Repository Documentation	40	Dev1	DT, OS, HW, GIT, AZR, VSC, WEB
Frontend and Backend Development		128		
ER	Expense and Report Development	64		
ER1	Expense Uploads	16	Dev1	DT, OS, HW, GIT, AZR, VSC, WEB
ER2	Uploaded Expenses Real-time Dashboard	4	ER1, BD6	DT, OS, HW, GIT, AZR, VSC, WEB
ER3	AI-Extracted Expense Data Review and Correction	20	ER2	DT, OS, HW, GIT, AZR, VSC, WEB
ER4	Uploaded Expenses Deletion	4	ER2	DT, OS, HW, GIT, AZR, VSC, WEB
ER5	Expense Reports View, Creation, Edit, Submission, and Deletion	20	ER3, ER4	DT, OS, HW, GIT, AZR, VSC, WEB
UA	User Access Development	64		

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ID	Name	Time (h)	Dependencies	Resources
UA1	Company Email User Authentication	8	BD7	DT, OS, HW, GIT, AZR, VSC, WEB
UA2	Role-based User Access	16	UA1	DT, OS, HW, GIT, AZR, VSC, WEB
UA3	Submitted Report Approval and Rejection	8	ER5, UA2	DT, OS, HW, GIT, AZR, VSC, WEB
UA4	Employee Spending Limit Configuration	8	UA2	DT, OS, HW, GIT, AZR, VSC, WEB
UA5	Submitted and Reviewed Expenses Record Visualization	4	UA3	DT, OS, HW, GIT, AZR, VSC, WEB
UA6	Email Notification Service Integration	20	UA3	DT, OS, HW, GIT, AZR, VSC, WEB
BD	Backend Development	120		
BD1	RESTful API Backend Logic Implementation	20	Dev1	DT, OS, HW, GIT, AZR, VSC, WEB
BD2	Azure SQL Database Integration and Manipulation	20	BD1	DT, OS, HW, GIT, AZR, VSC, WEB
BD3	Azure Blob Storage Integration	8	BD2	DT, OS, HW, GIT, AZR, VSC, WEB
BD4	Microsoft Document Intelligence AI Model Integration	20	BD2, BD3	DT, OS, HW, GIT, AZR, VSC, WEB
BD5	Automated AI-driven Data Extraction Pipeline	20	BD4	DT, OS, HW, GIT, AZR, VSC, WEB
BD6	Azure SignalR Real-time Notification Service Integration	16	BD5	DT, OS, HW, GIT, AZR, VSC, WEB
BD7	Microsoft Entra ID User Authentication Integration	16	BD1	DT, OS, HW, GIT, AZR, VSC, WEB
FD	Frontend Development	10		
FD1	Expenses Search and Filter	8	ER2	DT, OS, HW, GIT, AZR, VSC, WEB
FD2	Reports Search	2	ER5	DT, OS, HW, GIT, AZR, VSC, WEB
Total		513		

Table 5.3.1: Summary of project tasks with effort, dependencies, and resources. [Own creation]

5.4 Gantt Diagram

A Gantt diagram is attached at the end of this section. Continuous arrows represent Finish-Start (FS) dependencies between tasks, while dashed arrows represent Finish-Finish (FF) dependencies. FS dependencies indicate that successor tasks cannot start until their predecessor tasks are completed. FF dependencies show that successor tasks cannot finish until their predecessor tasks are completed.

Referring to Diagram 5.4.1, the project timeline spans **122 days** with a **one-week built-in buffer**. Considering the 513 total estimated hours, this equates to approximately **4 hours of work per day**. Consequently, the weekly workload is 28 hours, matching the weekly effort allocated in the Gantt diagram. Hence, the task effort estimations from Table 5.3.1 are translated into the Gantt diagram using this pattern (e.g., the ER1 task, which costs 20 hours, represents 4 days of work in the diagram).

Since the Agile Methodology is selected for this application development, each week in the Gantt diagram represents one sprint. To perform the Agile ceremonies described in Section 4.1, daily Scrum (15 minutes each) is scheduled every 2 days, resulting in 3 meetings per week that happen on every Monday, Wednesday, and Friday. Sprint planning, review, and retrospective require 45 minutes each. Thus, Agile ceremonies total 3 hours per week, which equals 51 hours over 17 weeks. Agile's iterative meeting structure spans the **PM1** task across the entire Gantt diagram timeline. However, this task **should be represented iteratively**, since the Gantt diagram tool does not support this behaviour, it is represented as a concurrent task.

Other project management tasks follow the same pattern:

- **PM2** remains undetermined due to undefined thesis director meetings, representing iterative appointments throughout the project lifetime.
- **PM3** starts after **PS2** (temporal planning) is completed and runs concurrently with full application development, as the product backlog requires constant updates whenever new features or tasks are added.
- **FPD2** runs over the project stage to ensure consistency with final deliverables.
- **Dev5**, which depends on **Dev1** (beginning once the repository is created), spans most of the project duration to satisfy objective SG7 from Section 3.1, as clean code requires ongoing, extensive documentation.

Certain development tasks also demand additional time. These tasks have an initial effort estimation of **8 hours (2 days of the Gantt diagram)**. Since these tasks face a high probability of experiencing significant challenges, the **remaining effort time** is allocated as a **buffer**.

Notable examples include **ER1**, **ER3**, **UA2**, and **UA6**. **ER1**. **ER1** is the first coding task; **ER3** requires extensive coordination with frontend and backend; **UA2** requires a prior database record of the employee hierarchy; **UA6** marks the author's first experience implementing email notifications. Finally, **Backend Development** tasks similarly incur additional time costs.

Expenses Reporting Tool for an IT Consultancy

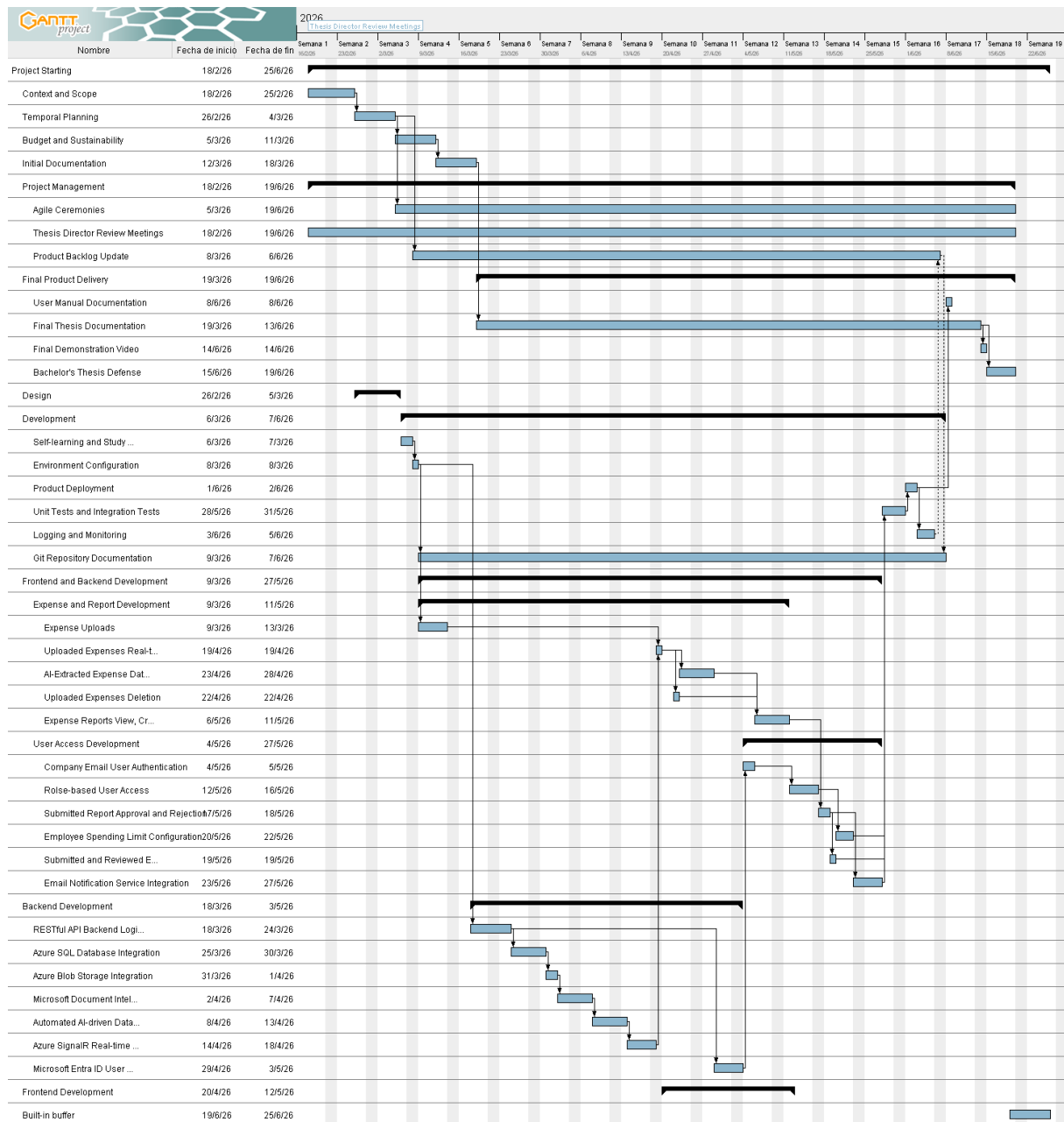


Figure 5.4.1: Gantt Diagram. [own creation]

5.5 Risk Management

Possible risks affecting project progress are outlined in Section 3.3. With the timeline now established, contingency plans are defined to mitigate potential obstacles. This section presents each identified risk along with its probability, impact, and corresponding mitigation strategy. Moreover, all identified risks, along with the corresponding affected tasks and the contingency resources required for mitigation, are detailed in Table 5.5.1.

Coding Bugs

Probability: Medium

Impact: High

Coding bugs are inevitable, making thorough testing essential. As outlined in Section 5.4, every code change undergoes rigorous testing across all use cases, including edge cases. During the Dev1 task, ESLint and SonarQube Visual Studio Code extensions are installed to automate bug detection from the very beginning of the implementation.

If bugs persist, additional debugging time is allocated, using **10 hours** of the buffer time. It is important to fix critical bugs before proceeding with new feature development, prioritizing functional application delivery.

However, thanks to the installed extensions and consistent testing, the number of bugs is minimized, reducing the likelihood of significant setbacks. Furthermore, complex tasks already include built-in buffer time, as mentioned in section 5.4, and the reserve hours are available to cover any delays that may arise.

Tight Project Deadlines

Probability: High

Impact: Medium

Software development projects often struggle with tight deadlines. Due to the author's limited professional experience, deviation between the actual task effort and the initial time estimation is anticipated. However, since this project follows Agile methodology, iterative meetings help to detect and manage overruns at an early stage.

To overcome this challenge, the remaining **9 hours** of overhead serve as a lifesaver. This is an intentional allocation, as the project is scheduled to finish several days before the thesis defense period. Therefore, a one-week delay is permitted without affecting the final submission.

In the worst-case scenario, if the project misses the expected deadline, certain non-essential application features will be deferred to the final phase, depending on the time available. Nevertheless, this alternative can only be made after obtaining the Product Owner's approval.

Lack of Experience in Certain Technologies

Probability: High

Impact: Medium

Given that several technologies used in application implementation lie beyond the author's expertise, mitigation plans have been established to address this challenge. Task Dev0 specifically accounts for this difficulty by allocating dedicated time for self-study, thereby minimizing potential setbacks.

Since a prior study cannot fully mitigate this risk, a proposed solution is to set aside up to **8 hours** from the available buffer. If delays are prolonged, assistance from the FSP team serves as a fallback. Seeking timely assistance is an essential professional skill, as it helps prevent unnecessary deadline extensions.

Company's Licenses and Security Permissions

Probability: Medium

Impact: Low

Another risk involves FSP's security system. Cyberattacks are taken more seriously every day, often requiring developers to request and wait for permission to install resources and extensions. Consequently, new restrictions may limit initial resource availability.

To prevent this blocker, Dev1 has been created to install all the previously identified tools and extensions in advance. However, this prior setup does not remove the probability of installing forthcoming tools.

As a contingency, a reordering of the project backlog can be considered using the Gantt chart (Figure 5.4.1) as a reference. Tasks with no dependencies, such as BD7 and the subsequent UA category tasks, may be prioritized and initialized earlier. Likewise, other non-dependent tasks (e.g., Dev5 task) may also be advanced to maintain project progress.

Risk	Probability	Time (h)	Affected Tasks	Resources
Coding Bugs	15%	10	Dev3, ER1-5, UA1-6, BD1-7, FD1-2	DT, HW
Tight Project Deadlines	30%	9	Dev2-4, ER1-5, UA1-6, BD1-7, FD1-2	DT, HW
Lack of Experience in Certain Technologies	30%	8	Dev2, Dev3, ER1-5, UA1-6, BD1-7,	DT, HW
Company's Licenses and Security Permissions	15%	0	Dev2, BD1-7	DT, HW

Table 5.5.1: Summary of identified risks, their probabilities, affected tasks, and contingency duration and resources. [Own creation]

6 Budget and Sustainability

The final fundamental aspect for concluding the initial documentation of a project is the budget and the project's economic viability. In addition, it is important to analyse the project's sustainability, considering the impact of the proposed solution on economic, environmental, and social dimensions. This stage studies both positive and negative project consequences.

6.1 Budget

This section details all project expenses, justifying each cost and its corresponding calculation. The ultimate goal is to determine the initial project budget.

6.1.1 Personnel Cost

To obtain realistic personnel cost estimates, the average hourly wage for each personnel member was calculated in advance using market salary data [1]. This calculation assumes 260 working days per year with an average of 7.5 working hours per day.

In Table 6.1.1, the hourly wage calculation for each member of personnel is specified. Note that the final number already includes the Social Security (SS) cost. Also note that personnel are categorized by roles rather than by the human resources (HR) identified in Section 5.2.1. Each resource is assigned to at least one role in the table. There are five different roles involved in this project:

- **Project Manager (PM)**: Responsible for planning and leading the entire project.
- **Junior Full-Stack Developer (D)**: Responsible for the full coding implementation of the application
- **QA Tester (T)**: Responsible for verifying the applications' functionality.
- **Software Architect (SA)**: Responsible for designing the application's technical structure while considering the project requirements and scope.
- **Technical Writer (TW)**: Responsible for elaborating a clear and accurate project documentation.

Role	Annual Salary (€)	Total including SS (€)	Cost / Hour (€)	Assigned HR
Project Manager (PM)	43,300	56,290	28.87	SM, PO, TD, GT
Full-stack Junior Developer (D)	29,800	38,740	19.87	DT
QA Tester (T)	26,888	34,954.4	17.93	DT
Software Architect (SA)	41,800	54,340	27.87	DT
Technical Writer (TW)	31,300	40,690	20.87	DT

Table 6.1.1: Annual and hourly salaries for different project roles. [Own creation]

In Table 6.1.2, personnel costs for each task and role, as well as the overall cost, are calculated. These costs result from multiplying each personnel member's effort hours (from Table 5.3.1) by their corresponding role's hourly wage (from Table 6.1.1).

ID	Name	Total hours	Hours per role					Cost (€)
			PM	D	T	SA	TW	
PS	Project Starting	50	80	0	0	0	0	1,433.33
PS1	Context and Scope	20	20	0	0	0	0	577.33
PS2	Temporal Planning	10	10	0	0	0	0	288.67
PS3	Budget and Sustainability	10	10	0	0	0	0	288.67
PS4	Initial Documentation	10	10	0	0	0	0	288.67
PM	Project Management	67	59	67	0	0	0	3,034.20
PM1	Agile Ceremonies	51	51	51	0	0	0	2,485.40
PM2	Thesis Director Review Meetings	8	8	8	0	0	0	389.87
PM3	Product Backlog Update	8	0	8	0	0	0	158.93
DT	Design	16	0	0	0	16	0	445.87
DT1	Use Case Diagram	2	0	0	0	2	0	55.73
DT2	Conceptual Model Diagram	4	0	0	0	4	0	111.47
DT3	Technology Stack Decision	2	0	0	0	2	0	55.73
DT4	Logic and Physic Architecture Design	4	0	0	0	4	0	111.47
DT5	UI mockups Design	2	0	0	0	2	0	55.73
DT6	Security Design	2	0	0	0	2	0	55.73
FPD	Final Product Delivery	46	0	5	0	0	41	954.87
FPD1	User Manual Documentation	2	0	1	0	0	1	40.73
FPD2	Final Thesis Documentation	20	0	0	0	0	20	417.33
FPD3	Final Demonstration Video	4	0	4	0	0	0	79.47
FPD4	Bachelor's Thesis Defense	20	0	0	0	0	20	417.33
Dev	Development	76	0	71	5	0	0	1500.16
Dev0	Self-learning and study of unfamiliar technologies	8	0	8	0	0	0	158.93
Dev1	Environment Configuration	2	0	2	0	0	0	39.73
Dev2	Product Deployment	8	0	8	0	0	0	158.93
Dev3	Unit and Integration Testing	10	0	5	5	0	0	188.96
Dev4	Logging and Monitoring	8	0	8	0	0	0	158.93
Dev5	Git Repository Documentation	40	0	40	0	0	0	794.67
Frontend and Backend Development		128	0	128	0	0	0	2,542.93
ER	Expense and Report Development	64	0	64	0	0	0	1,271.47
ER1	Expense Uploads	16	0	16	0	0	0	317.87
ER2	Uploaded Expenses Real-time Dashboard	4	0	4	0	0	0	79.47
ER3	AI-Extracted Expense Data Review and Correction	20	0	20	0	0	0	397.33
ER4	Uploaded Expenses Deletion	4	0	4	0	0	0	79.47
ER5	Expense Reports View, Creation, Edit, Submission, and Deletion	20	0	20	0	0	0	397.33

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ID	Name	Total hours	PM	Hours per role				Cost (€)
				D	T	SA	TW	
UA	User Access Development	64	0	64	0	0	0	1,271.47
UA1	Company Email User Authentication	8	0	8	0	0	0	158.93
UA2	Role-based User Access	16	0	16	0	0	0	317.87
UA3	Submitted Report Approval and Rejection	8	0	8	0	0	0	158.93
UA4	Employee Spending Limit Configuration	8	0	8	0	0	0	158.93
UA5	Submitted and Reviewed Expenses Record Visualization	4	0	4	0	0	0	79.47
UA6	Email Notification Service Integration	20	0	20	0	0	0	397.33
BD	Backend Development	120	0	120	0	0	0	2,384.00
BD1	RESTful API Backend Logic Implementation	20	0	20	0	0	0	397.33
BD2	Azure SQL Database Integration and Manipulation	20	0	20	0	0	0	397.33
BD3	Azure Blob Storage Integration	8	0	8	0	0	0	158.93
BD4	Microsoft Document Intelligence AI Model Integration	20	0	20	0	0	0	397.33
BD5	Automated AI-driven Data Extraction Pipeline	20	0	20	0	0	0	397.33
BD6	Azure SignalR Real-time Notification Service Integration	16	0	16	0	0	0	317.87
BD7	Microsoft Entra ID User Authentication Integration	16	0	16	0	0	0	317.87
FD	Frontend Development	10	0	10	0	0	0	198.67
FD1	Expenses Search and Filter	8	0	8	0	0	0	158.93
FD2	Reports Search	2	0	2	0	0	0	39.73
Total		513	109	401	5	16	41	12,504.03

Table 6.1.2: Personnel cost breakdown by task, role, and total personnel cost. [Own creation]

6.1.2 Generic Cost

In addition to personnel costs, other project costs must also be considered. Therefore, the missing costs are detailed in this section, followed by the formula applied to obtain the resulting cost.

Amortization

Physical resources are also required and used in the project. However, considering only the product purchase price would not be entirely correct. For this reason, the depreciation of each hardware resource has been calculated.

Hardware resources have a 4-year depreciation period. Other relevant factors for amortization calculation include the total project duration and the number of hours dedicated to the project each day, respectively 122 days and 4 hours per day. Since the hardware resources are used throughout the project timeline, their total usage time corresponds to the total project hours (513 hours). Accordingly, the subsequent equation calculates the amortization costs for each hardware resource shown in Table 6.1.3:

$$\text{Amortization (€)} = \text{Resource price} \times \frac{1}{4 \text{ years}} \times \frac{1}{122 \text{ days}} \times \frac{1}{4 \text{ hours}} \times \text{hours used}$$

Hardware Resources	Price (€)	Quantity	Total Price (€)	Usage Time (h)	Amortization (€)
PC Dell 16 Pro [2]	1,193.53	1	1,193.53	513	313.67
Wireless Mouse [3]	11.99	1	16.99	513	4.47
Dell Monitor [4]	171.82	2	343.64	513	90.31
Keyboard [5]	19.99	1	19.99	513	5.25
Total					408.44

Table 6.1.3: Amortization cost of the project's hardware resources. [Own creation]

Electric cost

The current cost of electricity is around **0.1748 €/kWh** [6]. Electricity costs are calculated based solely on the electricity-consuming hardware resources and their respective usage times. Following the same approach used for amortization cost calculation, the usage time is **513 hours**. Table 6.1.4 details the calculations and the resulting total cost:

Hardware Resources	Power (W)	Usage Time (h)	Energy Consumed (kWh)	Electricity Cost (€)
PC Dell 16 Pro [2]	65	513	33.345	5.83
Wireless Mouse [3]	0.05	513	0.02565	0.00
Dell Monitor [4]	24	513	12.312	2.15
Keyboard [5]	0.1	513	0.0513	0.01
Total				7.99

Table 6.1.4: Electricity cost of the project's hardware resources. [Own creation]

Internet cost

The basic internet plan costs €21.95 per month [7]. Since the project lasts 122 days, the internet cost is calculated as follows:

$$\text{Internet cost} = 21.95 \text{ €/month} \times \frac{1 \text{ month}}{30 \text{ days}} \times 122 \text{ days} = \mathbf{€89.26}$$

Travel cost

The public transport rate in Barcelona is 45.5€ per 3 months [8]; the total travel cost corresponds to the following equation:

$$\text{Travel cost} = 45.5 \text{ €/3 months} \times \frac{3 \text{ months}}{90 \text{ days}} \times 122 \text{ days} = \mathbf{€61.88}$$

Office rental cost

The project development always takes place in the FSP office. Therefore, this is the only cost to consider in this category. The FSP office is located in Plaça Francesc Macià, Barcelona. The monthly rental fee is €6,000. Given that there is enough space to accommodate up to 25 employees working at the same time, the total cost is calculated as follows:

$$\text{Office rental cost} = 6,000 \text{ €/month} \times \frac{1 \text{ month}}{30 \text{ days}} \times \frac{1 \text{ office}}{25 \text{ employees}} \times 122 \text{ days} = \mathbf{€976.00}$$

Software resources cost

Finally, the cost of software resources is also considered. However, most of them are open-source and therefore excluded from the total cost, as they are available for free. Only resources that require payment are listed below. Regarding Azure resources, only those that incur a cost are included, since Azure offers a pay-as-you-go fee [9]. See Table 6.1.5 for a detailed breakdown of the cost of Azure resources.

Azure resources	Cost (€/month)
SQL DataBase	3.90
Bandwidth	0.01
Blob Storage	0.01
Log Analytics	0.01
Event Grid	0.01
Service Bus	0.01
Functions	0.01
Total	3.96

Table 6.1.5: Azure resource cost. [Own creation]

Given that the Azure resource cost is €3.96 per month, the total cost for the project (122 days) is calculated as follows:

$$\text{Azure resource cost} = 3.96 \text{ €/month} \times \frac{1 \text{ month}}{30 \text{ days}} \times 122 \text{ days} = \mathbf{€24.16}$$

Microsoft is also included in the software resources cost, demanding an annual fee of €99 [10]. This resource cost is calculated in the next equation:

$$\text{Microsoft resource cost} = 99 \text{ €/year} \times \frac{1 \text{ year}}{365 \text{ days}} \times 122 \text{ days} = \mathbf{€33.09}$$

Thus, the total software resources cost equals the sum of both costs:

$$€24.16 + €33.09 = \mathbf{€57.25}$$

Total generic cost

By summing all previously calculated costs, the total generic cost is obtained, as shown in Table 6.1.6.

Concept	Cost (€)
Amortization	408.44
Electricity	7.99
Internet	89.26
Travel	61.68
Office Rental	976.00
Software	57.25
Generic Cost (GC)	1,600.63

Table 6.1.6: Total generic cost. [Own creation]

6.1.3 Other costs

Contingencies

No project strategy can eliminate the possibility of encountering obstacles. Therefore, a time buffer (27 hours) has been included to cover potential setbacks. However, simply adding extra time is insufficient, as it also entails additional resource consumption. Consequently, a portion of the project budget must be allocated for these potential scenarios. The proportional buffer time allocated to the total project duration is 5% (27h/420). Therefore, the contingency cost amounts to **€705.23**, which is the 5% of the total general cost calculated so far (PC (€) + GC (€) = 13,370.03 + 1,600.92 = €14,970.94).

Incidental Costs

Incidental costs encompass the project risks outlined in Section 5.5. The cost of each risk is calculated by considering its probability, time cost, and required resources. Table 6.1.7 details these calculations. Note that the resources required for each risk are identified as Rx (R1, R2, ...).

Incident	Risk (%)	Time (h)	R1	R1 Cost (€/h)	R2	R2 Cost (€/h)	Total Cost (€)	Risk Cost (€)
Coding Bugs	15	10	D	19.87	HW	7.96	206.63	30.99
Tight Project Deadlines	30	9	D	19.87	HW	7.17	185.97	55.79
Lack of Experience in Certain Technologies	30	8	D	19.87	HW	6.37	165.30	49.59
Company's Licenses and Security Permissions	15	0	D	19.87	HW	0.00	0.00	0.00
Total								136.37

Table 6.1.7: Incidental cost. [Own creation]

6.1.4 Total cost

After summing the total costs from each section, the final project budget is determined, as shown in Table 6.1.8.

Concept	Cost (€)
Personnel Cost (PC)	12,504.03
Generic Cost (GC)	1,600.63
Contingency Cost (CC)	705.23
Incidental Cost (IC)	136.37
Total Budget	14,946.26

Table 6.1.8: Total cost of the project. [Own creation]

6.1.5 Management Control

Challenges, both expected and unexpected, are always present in any project. Even with a precisely estimated timeline and budget, it is very common for a project to fall short of the initial plan. To monitor costs during project execution, various deviation formulas will be applied in the three areas:

- **Human resources deviation:** To contrast the actual effort with the previous estimated effort of human resources in each task.

$$\text{Human resources deviation} = \sum_{i \in pit} (\text{estimated_cost_per_hour}_i - \text{real_cost_per_hour}_i) \times \text{total_real_hours}_i$$

Where i indexes each individual in pit , the set of all personnel involved in that task.

- **Amortization deviation:** To compare the previously estimated use of hardware resources with the actual usage time of those resources.

$$\text{Amortization deviation} = \sum_{i \in hr} (\text{estimated_usage_hours}_i - \text{real_usage_hours}_i) \times \text{price_per_hour}_i$$

Where i indexes each resource in hr , the set of all hardware resources used in the project.

- **Total cost deviation:** To determine each task's real cost deviation from the estimated cost. The general cost described in the formula represents the sum of the personnel and generic costs, excluding contingency and incident costs. Hence, the following formula is applied to each task:

$$\text{Total cost deviation} = (\text{estimated_general_cost} - \text{real_general_cost})$$

All cost deviation calculations require the real project costs across different categories. Before applying the deviation formulas, the real costs are calculated at the end of the project using the same methodology as the initial budget estimation.

Note that the resulting total cost deviation value can be either positive or negative. A positive value indicates an overestimation of the initial budget for that resource or task. In this case, the additional budget can be reallocated to address issues that arise during the project. Conversely, a negative value indicates an underestimation, requiring contingency and incident costs to cover the budget shortfall.

6.2 Sustainability

The technological revolution has accelerated dramatically over the last few decades. A clear example is the launch of the first telephone (1876), the first computer (1937), the first laptop (1974), the first smartphone (2007), and the first artificial intelligence (2017) [11]. This trend reveals that the time it takes for a technological invention to emerge and significantly impact society continues to shrink. This exponential growth underscores the increasing importance of evaluating the economic, environmental, and social impacts of technological projects. These three dimensions together constitute sustainability, the topic discussed in this section.

Technology is shaping our future, at least in the coming years. The primary mission of every engineer is to make people's daily lives easier. However, as mentioned earlier, developing a solution is not enough; a study of its sustainability effects is also a must. Therefore, addressing sustainability questions enables a better evaluation of the project's impact across the economy, environment, and society. This analysis raises awareness of the weight of sustainability in project development.

The author's sustainability consciousness is quite limited. The knowledge extends only to well-known aspects such as environmental footprint and social impact. The economic dimension and multiple sustainability indicators require further study. The subsequent subsection answers a series of key sustainability questions within the author's current expertise.

6.2.1 Economic Impact

Regarding PPP: Have you estimated the project execution cost?

The project budget is detailed in Section 6.1. All required costs have been considered, including personnel, hardware, software, contingencies, and incidents. Only indispensable resources are used and are included in the budget.

The computer used throughout the project was recently purchased and remains under warranty. Since there is an overabundance of hardware resources in the office, they can be easily replaced if needed. Regarding the software resources, they were already available to the company and are virtual; as a consequence, fixes and repairs are not included in the budget.

Additionally, only essential tasks demand a higher time cost. Since the AI model is connected to an external resource, this development phase can be significantly reduced while obtaining accurate data extraction. From an objective standpoint, a meticulous budget estimate has been established.

Regarding Useful Life: How are currently solved economic issues (costs...) related to the problem that you want to address (state of the art)?

The current expense management process imposes an excessive time cost for employees, who manually enter expenses into Excel spreadsheets. The same challenge is encountered in the finance department, which must verify the accuracy of each line item before approving reimbursements. Additionally, the current process is prone to human error regarding manual expense data entry and review.

In short, this time-consuming, low-efficiency process delivers minimal value while misallocating company personnel costs for both employees and finance staff.

Regarding Useful Life: How will your solution improve economic issues (costs, etc.) with respect to other existing solutions?

The proposed solution automates the current cumbersome process. It provides automatic receipt data extraction and a more intuitive dashboard for displaying receipt information. This eliminates the tedious manual data entry, enhances visualization beyond traditional spreadsheets, and reduces human error. As a result, the workflow is significantly streamlined, delivering accurate receipt processing while maximizing efficiency. This enables employees to focus on higher-value tasks, thus maximizing personnel investment returns.

6.2.2 Environmental Impact

Regarding PPP: Have you estimated the project's environmental impact?

The only resource directly affecting the environmental footprint is energy consumption. This resource is required throughout the entire project, as specified in Section 6.1.2. Both hardware resources and the shared workspace consume electricity continuously. However, hardware resource consumption is vital for application development and sharing the workspace's lighting with other employees reduces its environmental impact.

Regarding PPP: Have you considered minimizing its impact, for example, by reusing resources?

The application uses Azure for software resources. Currently, the Azure resources required for application development cost less than traditional physical resources. This cost optimization results from significantly reduced resource consumption, as cloud resources eliminate the need for transportation and frequent physical replacements. Updates and fixes are performed virtually, further reducing hardware requirements and electronic waste.

Furthermore, training the AI model requires substantial energy consumption. Reusing existing solutions, as decided in this project, optimizes this energy-intensive development process. Hence, the entire application development process prevents excessive resource use. As a result, tasks can be significantly reduced while avoiding excessive time costs.

Regarding Useful Life: How is the problem that you want to address currently solved (state of the art)?

The extensive time employees and finance staff dedicate to Excel spreadsheets results in excessive energy use. Switching the personnel to process expenses in the application also requires an energy-saving approach.

The primary strategy for reducing resource consumption is to develop the most efficient application possible, avoiding unnecessary resources and inefficient code that could overuse resources. This ensures that only essential resources are consumed each time employees use the application, limiting overall resource consumption throughout its lifecycle. Therefore, writing high-quality code is an essential skill for a programmer and is strongly considered throughout the project development.

Regarding Useful Life: How will your solution improve the environment with respect to other existing solutions?

This application is tailored for the company's exclusive use and excludes any irrelevant functionality from the final product. It streamlines resource consumption by avoiding the unnecessary computation performed by other existing applications.

6.2.3 Social Impact

Regarding PPP: What do you think you will achieve—in terms of personal growth—from doing this project?

This project enables the author to apply prior academic knowledge and software development skills in a real-world context. As a result, it enriches their working experience as future software engineer. Furthermore, it is the author's first experience managing and leading an entire project independently, strengthening self-learning, self-management skills and the ability to build software projects from scratch.

Regarding Useful Life: How is the problem that you want to address currently solved (state of the art)?

The current expense management process is handled manually through traditional methods. This involves manually entering data into Excel spreadsheets, emailing them to the finance department, and reviewing them line by line.

Regarding Useful Life: How will your solution improve the quality of life (social dimension) with respect to other existing solutions?

The proposed solution optimizes time spent on expense management while reducing employees' frustration and fatigue. This frees employees from this tedious task, allowing additional time to focus on more meaningful work. As a result, it maximizes employees' productivity.

Regarding Useful Life: Is there a real need for the project?

Regarding all project sustainability impacts (social, environmental, and economic), there is a clear need to build a robust solution. This solution overcomes challenges from all involved parties while introducing a sustainability-focused approach.

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